Modularity criteria for multiple Eisenstein series

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For $\tau \in \mathbb{H}$ put $q = e^{2\pi i \tau}$. For $\Bbbk = (k_1, \dots, k_r)$ with $k_1, \dots, k_r \geq 2$ and weight $k = k_1 + \cdots + k_r$,

$$G_{\Bbbk}(\tau) = \sum_{0 \prec \lambda_1 \prec \cdots \prec \lambda_r} \frac{1}{\lambda_1^{k_1} \cdots \lambda_r^{k_r}} = \zeta(k_1, \dots, k_r) + \sum_{n \geq 1} a_n q^n, \qquad a_n \in \mathbb{C}.$$

The index $\Bbbk$ is **admissible** if $k_r \geq 3$. Then the series defining $G_{\Bbbk}$ converges absolutely.

Example In weight 6:

$$G_{3,3} = \zeta(3,3) + \Big(\frac{4\pi^6}{15} + 4\zeta(3)\pi^3 i\Big)q + \Big(\frac{4\pi^6}{5} + 20\zeta(3)\pi^3 i\Big)q^2 + \cdots,$$

$$G_{2,4} = \zeta(2,4) + \Big(\frac{32\pi^6}{45} + 8\zeta(3)\pi^3 i\Big)q + \Big(\frac{112\pi^6}{15} + 40\zeta(3)\pi^3 i\Big)q^2 + \cdots.$$

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One can check that $2G_{3,3} - G_{2,4} = G_6$ and $2G_{3,3} - G_{2,4} \in \mathbb{R}[[q]]$.

Modular forms

Slash action

For $F\colon \mathbb{H} \to \mathbb{C}$, $k \in \mathbb{Z}$ and $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ put

$$(F|_k\gamma)(\tau) = (c\tau + d)^{-k} F\Big(\frac{a\tau + b}{c\tau + d}\Big).$$

The group $\mathrm{SL}_2(\mathbb{Z})$ is generated by $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ and $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$.

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A modular form of weight k for $\mathrm{SL}_2(\mathbb{Z})$ is a holomorphic function $F\colon \mathbb{H} \to \mathbb{C}$ with $F = \sum_{n \geq 0} a_n q^n$ and

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Question

Which linear combinations of MES are modular forms?

Main result

Let M_k be the space of modular forms of weight k for $\mathrm{SL}_2(\mathbb{Z})$. Let $\mathbb{k}_1, \dots, \mathbb{k}_N$ be all indices $(k_1, \dots, k_r)$ with $k_r \geq 3$ and $k_1 + \cdots + k_r = k$. For $\mathbb{A} = \mathbb{R}$ or $\mathbb{Q}$ put

$$\mathfrak{E}_k^{\mathrm{adm}} = \mathrm{span}_{\mathbb{C}}\{G_{\mathbb{k}_1}, \dots, G_{\mathbb{k}_N}\}, \qquad \mathfrak{E}_k^{\mathbb{A},\mathrm{adm}} = \mathfrak{E}_k^{\mathrm{adm}} \cap \mathbb{A}[[q]], \qquad M_k^{\mathbb{A}} = M_k \cap \mathbb{A}[[q]].$$

Theorem (*Y., 2026*)

For every $k \geq 1$,

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Let $F = \sum_{i=1}^{N} c_i G_{\mathbb{k}_i}$ with $c_1, \dots, c_N \in \mathbb{C}$.

- If $F \in \mathbb{R}[[q]]$, then $F \in M_k$. For odd k this gives $F = 0$.
- If k is even and $c_1, \dots, c_N \in \mathbb{R}$, then $F \in M_k \iff F \in \mathbb{R}[[q]]$.

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Corollary

(i) The $\mathbb{C}$-vector space $\left\{(c_1, \dots, c_N) \in \mathbb{C}^N \;\middle|\; \sum_{i=1}^N c_i G_{\mathbb{k}_i} \in M_k\right\}$ has a basis in $\mathbb{Q}^N$.

(ii) If $c_1, \dots, c_N \in \mathbb{Q}$, then $F \in M_k \iff (2\pi i)^{-k} F \in \mathbb{R}[[q]] \iff (2\pi i)^{-k} F \in \mathbb{Q}[[q]]$.

- **Fixed weight:** $i\,(G_3 - \zeta(3))$ has real Fourier coefficients, but is not modular.
- **Admissible:** $G_{2,2}$ has real Fourier coefficients, but it is not modular, since $M_4 = \mathbb{C}G_4$ and

$$G_{2,2} = \frac{\pi^4}{120} - 2\pi^4 q + \cdots, \qquad G_4 = \frac{\pi^4}{90} + \frac{8\pi^4}{3} q + \cdots.$$

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Question

Allow $k_r = 2$. Let $\mathfrak{E}_k = \mathrm{span}_{\mathbb{C}}\{G_{k_1,\ldots,k_r} \mid k_1 + \cdots + k_r = k\}$. It contains the quasimodular forms $\widetilde{M}_k = \mathrm{span}_{\mathbb{C}}\{G_2^a G_4^b G_6^c \mid 2a + 4b + 6c = k\}$. Is $\mathfrak{E}_k^{\mathbb{Q}} = \widetilde{M}_k^{\mathbb{Q}}$?

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Answer: No. The space $\mathfrak{E}_k^{\mathbb{Q}}$ is larger than $\widetilde{M}_k^{\mathbb{Q}}$.
For example, in weight 8,

$$F_8 = 3G_{2,2,4} - 6G_{2,3,3} - 2G_{2,4,2} - 4G_{3,2,3} - 8G_{3,3,2},$$

$$(2\pi i)^{-8} F_8 = -\frac{1}{3110400} - \frac{q}{405} - \frac{13q^2}{2160} + \frac{122q^3}{405} + \frac{10787q^4}{6480} + O(q^5) \in \mathbb{Q}[[q]].$$

No linear combination of G_2^4, $G_2^2 G_4$, $G_2 G_6$, G_4^2 has these five coefficients, so $F_8 \notin \widetilde{M}_8$. In fact

$$\mathfrak{E}_8^{\mathbb{Q}} = \widetilde{M}_8^{\mathbb{Q}} \oplus \mathbb{Q}\,(2\pi i)^{-8} F_8.$$

Sketch of the proof: transformation formula

Every $F \in \mathfrak{E}_k^{\mathrm{adm}}$ has $F = \sum_{n \geq 0} a_n q^n$, so $F|_k T = F$. Since S and T generate $\mathrm{SL}_2(\mathbb{Z})$, we have $F \in M_k \iff F|_k S = F$. So the question is how $F|_k S$ differs from F. The answer uses

$$F^\flat(\tau) = (-1)^k \, \overline{F(-\bar{\tau})} = \sum_{n \geq 0} (-1)^k \, \overline{a_n} \, q^n, \qquad \text{so for even } k\text{:} \quad F^\flat = F \iff \text{all } a_n \in \mathbb{R}.$$

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Let $F \in \mathrm{span}_{\mathbb{R}}\{G_{k_1,\dots,k_r} \mid k_1 + \cdots + k_r = k,\ k_r \geq 3\}$, *and let* $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ *have bottom row* (c,d) *with* $c \geq 1$. *Then*

$$F|_k\gamma - (-1)^k F = \sum_{c'\geq 1,\ \gcd(c',d')=1,\ d'/c' \leq d/c} (F - F^\flat)|_k \gamma_{c',d'}, \qquad \gamma_{c',d'} = \left(\begin{smallmatrix} * & * \\ c' & d' \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z}),$$

and the series on the right converges absolutely. In particular, for $\gamma = S$,

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For real coefficients c_i and even k: $\quad F \in \mathbb{R}[[q]] \ \Rightarrow\ F^\flat = F \ \Rightarrow\ F|_k S = F \ \Rightarrow\ F \in M_k$.

Sketch of the proof: complex coefficients

The transformation formula holds only for real c_i, since $\left(\sum_i c_i G_{\mathbb{k}_i}\right)^\flat = \sum_i \overline{c_i}\, G^\flat_{\mathbb{k}_i}$. How do we treat complex c_i?

Lemma (*Y., 2026*)

Let $F \in \mathrm{span}_{\mathbb{R}}\{G_{k_1,\dots,k_r} \mid k_1+\cdots+k_r = k\}$. *Then* $F^\flat - F \in \mathfrak{E}_0 + \mathfrak{E}_1 + \cdots + \mathfrak{E}_{k-1}$.

For example, in weight 6,

$$G^\flat_{3,3} - G_{3,3} = -2\zeta(3)\big(G_3 - \zeta(3)\big) \in \mathfrak{E}_0 + \mathfrak{E}_3.$$

Theorem (*Bachmann–Kanno, 2026*)

The sum $\mathfrak{E}_0 + \mathfrak{E}_1 + \mathfrak{E}_2 + \cdots$ *is direct.*

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Now let k be even and $F = \sum_i c_i G_{\Bbbk_i} \in \mathbb{R}[[q]]$ with $c_i = a_i + ib_i$, $a_i, b_i \in \mathbb{R}$. Put $A = \sum_i a_i G_{\Bbbk_i}$ and $B = \sum_i b_i G_{\Bbbk_i}$. Then

$$A + iB = F = F^\flat = A^\flat - iB^\flat \quad\Longrightarrow\quad 2iB = (A^\flat - A) - i(B^\flat - B) \in \mathfrak{E}_0 + \cdots + \mathfrak{E}_{k-1}.$$

Since $B \in \mathfrak{E}_k$, this gives $B = 0$. So F equals the real combination A, and $F \in M_k$ by the real case.

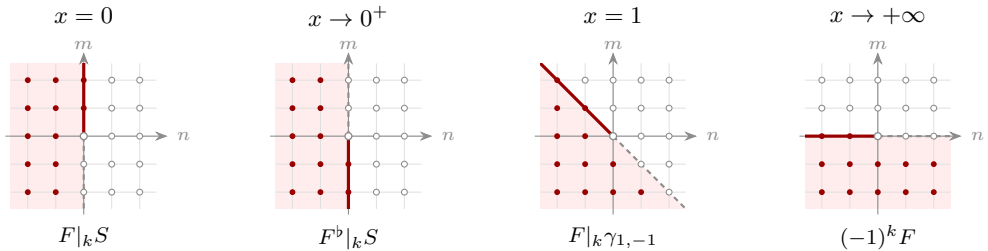
Thank you for your attention!

Appendix: proof of the transformation formula

Fix $\tau \in \mathbb{H}$. In the lattice sum $F(\tau)$ the positive points are $m\tau + n$ with $m > 0$, or $m = 0$ and $n > 0$. For $x \in \mathbb{Q}$ let $H(x)$ be the same lattice sum with the positive points

$$n + xm < 0, \quad \text{or} \quad n + xm = 0 \text{ and } m > 0 \qquad \text{(filled points below)}.$$

For example, $H(x) = \sum \frac{1}{(m\tau+n)^k}$ over these points for $F = G_k$. Bachmann (2012): $H(-d/c) = (F|_k\gamma_{c,d})(\tau)$.



As x grows, the half-plane rotates. A point changes sides only when the line passes through it. So H is a step function of x.

Appendix: the jumps

From $x = 0$ **to** $x \to 0^+$ (first two pictures) only the vertical axis flips. The points $j\tau$ with $j > 0$ leave, and those with $j < 0$ come in. For $F = G_k$,

$$H(0) - H(0^+) = \sum_{j>0} \frac{1}{(j\tau)^k} - \sum_{j<0} \frac{1}{(j\tau)^k} = \big(1 - (-1)^k\big) \frac{\zeta(k)}{\tau^k}.$$

At $x = u = -d/c$ the line through $c\tau + d$ flips in the same way. Flipping the horizontal line turns F into $F^\flat$, so

$$H(u) - H(u^+) = \big((F - F^\flat)|_k \gamma_{c,d}\big)(\tau), \qquad \text{e.g.} \quad \big(1 - (-1)^k\big) \frac{\zeta(k)}{(c\tau + d)^k} \quad \text{for } F = G_k.$$

All jumps from $x = 0$ to $x \to +\infty$, where $H(+\infty) = (-1)^k F$, add up to the transformation formula. For $F = G_k$,

$$G_k|_k S - (-1)^k G_k = \big(1 - (-1)^k\big)\zeta(k) \sum_{c \geq 1,\ d \leq 0,\ \gcd(c,d)=1} \frac{1}{(c\tau + d)^k}.$$

The main work is to show that the jumps are absolutely summable. Entries $k_i = 2$ need a cancellation.